(a) By time dilation, we know the microbes' proper time interval $\Delta\tau$ and the lab's time interval Δt are related by $\Delta t = \gamma\Delta\tau$, so

$$\Delta\tau = \Delta t/\gamma = \Delta t\sqrt{1-(v/c)^2} = \Delta t\sqrt{1-(\omega R)^2/c^2}, \tag{1}$$

or in other words,

$$1-\left(\frac{\omega R}{c}\right)^2 = \left(\frac{\Delta\tau}{\Delta t}\right)^2 \tag{2}$$

To find the angular velocity such that the microbes survive for 3 hours in their rest frame while the lab waits for 5 hours for the compound to arrive, we need to solve for ω using $\Delta\tau = 3$ and $\Delta t = 5$, we get

$$\omega = \frac{4c}{5R}. \tag{3}$$

(b) Since

$$d\tau = dt/\gamma = \sqrt{1-\frac{v^2(t)}{c^2}}dt, \tag{4}$$

the integral

$$\Delta\tau = \int_0^{t^*}\sqrt{1-\frac{v^2(t)}{c^2}}dt \tag{5}$$

tells us that when time t^* has passed in the lab frame, time $\Delta\tau$ has passed in the microbes' frame. Since we know the microbes die at $\Delta\tau = 3$ hours, we just need to solve for t^* to find out the time in the lab frame when the microbes die. Using $v(t) = v_0\sqrt{1-t/5}$, Eq. (5) becomes

$$\Delta\tau = \int_0^{t^*}\sqrt{1-\frac{v_0^2}{c^2}+\frac{v_0^2}{c^2}\frac{t}{5}}dt. \tag{6}$$

The calculation can be simplified if we define

$$\beta_0 \equiv \frac{v_0}{c} = \frac{4}{5}, \tag{7}$$

$$\gamma_0 \equiv \frac{1}{\sqrt{1-(v_0/c)^2}} = \frac{5}{3}, \tag{8}$$

and change the integration variable to $\rho = t/\Delta t$, where $\Delta t = 5$ (hours). Then Eq. (6) becomes

$$\frac{\Delta\tau}{\Delta t} = \int_0^{t^*/\Delta t}\sqrt{1-\beta_0^2+\beta_0^2\rho}d\rho \tag{9}$$

Using the integral formula provided on the equation sheet, this evaluates to

$$\begin{aligned}\frac{\Delta\tau}{\Delta t} &= \frac{2}{3\beta_0^2}[1+\beta_0^2(\rho-1)]^{3/2}\Big|_0^{t^*/\Delta t} \\ &= \frac{2}{3\beta_0^2}\left[\left(1-\beta_0^2+\beta_0^2\frac{t^*}{\Delta t}\right)^{3/2}-(1-\beta_0^2)^{3/2}\right].\end{aligned} \tag{10}$$

Using $\beta_0 = (1-\gamma_0^{-2})^{1/2}$ and simplify Eq. (10), we get

$$\frac{\Delta\tau}{\Delta t} = \frac{2}{3\gamma_0}\frac{1}{\gamma_0^2-1}\left\{\left[1+(\gamma_0^2-1)\frac{t^*}{\Delta t}\right]^{3/2}-1\right\}. \tag{11}$$

Recall that $\frac{\Delta\tau}{\Delta t} = \frac{1}{\gamma_0}$, we solve for t^* and get

$$t^* = \frac{1}{\gamma_0^2-1}\left[\left(\frac{3}{2}\gamma_0^2-\frac{1}{2}\right)^{2/3}-1\right]\Delta t = \frac{9}{16}\left[\left(\frac{11}{3}\right)^{2/3}-1\right]\Delta t = 3.88 \text{ hours}. \tag{12}$$

2) a) Recall $g_{ab} g^{bc} = \delta^a_c$

$$g_{ab} g^{ba} = g_{ab} g^{bc} \delta^a_c = \delta_a{}^c \delta^a_c = \delta^c_c = \sum_{\mu=0}^{D-1} 1 = D$$

where D is the spacetime dimension.

b) $\eta^{im} \eta_{mi} = \eta^{ij} \eta_{ji} = \delta^{ij} \delta_{ji} = 3$ (in 4 spacetime dims)

Not a Lorentz scalar.

c)
$$\begin{aligned}
\partial_\mu g^{\nu\rho} &= \partial_\mu (g^{\nu\kappa} g^{\rho\lambda} g_{\lambda\kappa}) \\
&= (\partial_\mu g^{\nu\kappa}) g_{\lambda\kappa} g^{\rho\lambda} + (\partial_\mu g^{\rho\lambda}) g^{\nu\kappa} g_{\lambda\kappa} + g^{\nu\kappa} g^{\rho\lambda} \partial_\mu g_{\lambda\kappa} \\
&= \delta^\rho_\kappa \partial_\mu g^{\nu\kappa} + (\partial_\mu g^{\rho\lambda}) \delta^\nu_\lambda + g^{\nu\kappa} g^{\rho\lambda} \partial_\mu g_{\lambda\kappa} \\
&= 2 \partial_\mu g^{\nu\rho} + g^{\nu\kappa} g^{\rho\lambda} \partial_\mu g_{\lambda\kappa}
\end{aligned}$$

$$\Rightarrow g^{\nu\kappa} g^{\rho\lambda} \partial_\mu g_{\lambda\kappa} = -\partial_\mu g^{\nu\rho}$$

d)
$$T^{(ab)} \omega_{[ab]} = T^{(ba)} \omega_{[ab]} = T^{(ab)} \omega_{[ba]} = -T^{(ab)} \omega_{[ab]}.$$

$$\Rightarrow T^{(ab)} \omega_{[ab]} = 0$$

e)
$$\begin{aligned}
\omega_{(ab)} u^a u^b &= \tfrac{1}{2}(\omega_{ab} + \omega_{ba}) u^a u^b \\
&= \tfrac{1}{2} \omega_{ab} u^a u^b + \tfrac{1}{2} \omega_{ba} u^a u^b \\
&= \tfrac{1}{2} \omega_{ab} u^a u^b + \tfrac{1}{2} \omega_{ab} u^b u^a \\
&= \omega_{ab} u^a u^b.
\end{aligned}$$

3) The rapidities are: $\varphi_u = \operatorname{arctanh}(u)$

$\varphi_v = \operatorname{arctanh}(v)$.

Since rapidities add, $\varphi_v = n\varphi_u$.

So $v = \tanh(\varphi_v) = \tanh(n\varphi_u)$.

(a)

$$A^{abc}g_{bd}S_{ce}g^{df}g^{h}{}_{a} = g^{h}{}_{a}A^{abc}g_{bd}g^{df}S_{ce} = A^{hbc}\delta_{b}{}^{f}S_{ce} = A^{hfc}S_{ce} \tag{20}$$

(b) The left hand side has only one free index a, but the right hand side has one extra free index d in addition to a.

(c) The trace of the metric is

$$g_{ab}g^{ab} = g_{ab}g^{ba} = g_{ab}g^{bc}\delta_{c}{}^{a} = \delta_{a}{}^{c}\delta_{c}{}^{a} = \delta_{a}{}^{a} = 4. \tag{21}$$

(d) Take the trace of the Einstein Equation, we get

$$\begin{aligned} g^{ab}\left(R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab}\right) &= 8\pi g^{ab}T_{ab} \\ R - 2R + 4\Lambda &= 8\pi T, \end{aligned} \tag{22}$$

where to get the second line, we have used the fact that the trace of metric is 4 in four spacetime dimensions. Rearrange the second line, we get

$$R = 4\Lambda - 8\pi T, \tag{23}$$

and plug back into the Einstein Equation,

$$\begin{aligned} R_{ab} &= 8\pi T_{ab} + \frac{1}{2}g_{ab}(4\Lambda - 8\pi T) - \Lambda g_{ab} \\ &= 8\pi T_{ab} + \Lambda g_{ab} - 4\pi T g_{ab} \\ &= 8\pi T_{ab} + g_{ab}(\Lambda - 4\pi T), \end{aligned} \tag{24}$$

which is the result that we want to show.